Jeremy Leipzig, PhD

BIOINFORMATICS ENGINEER & TECHNICAL PM

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SUMMARY

Technical product manager for TileDB's population-genomics line and a bioinformatics leader with 25+ years bridging genomics R&D and enterprise software. I develop scalable genomics data products from concept through launch — translating genetics research and discovery workflows into enterprise-grade platforms used by pharma and biotech partners, owning product vision, roadmaps, and success metrics while partnering across science, engineering, and business. I have helped four diagnostic, therapeutic, and SaaS startups define and implement their product strategy.

EXPERIENCE

Product Manager, TileDB | 2022 – Present

Manage the population genomics product line, including TileDB-VCF, end-to-end — product roadmapping, software development, sales demos, webinars, conference presentations, customer support, and marketing. Support 5 major biopharma and hospital customers, and have been involved in various stages of the software sales process for over 30 prospects. Built TileDB-specific solutions for GWAS and variant annotation at scale. Led Nextflow tooling for TileDB Carrara and ontologies for TileAI. Serve as the primary industry partner contact for the BeginNGS pediatric newborn screening consortium. Lead project management for teams of up to 7 developers.

Bioinformatics Lead, Truwl | 2020 – 2022

Onboarded tools, workflows, and high-impact analyses into the Truwl platform. Developed the benchmarking product and customer acquisition strategy.

Bioinformatics Engineer, Panorama Medicine | 2018 – 2019

Developed cloud-based pipelines and analysis for drug repositioning efforts. Lead for data mining and competitive intelligence research. First employee.

Senior Bioinformatics Scientist, CytoVas LLC | 2017 – 2018

Scaled up flow cytometry workflows for lab developed tests. Developed novel statistical analyses to measure subpopulations of cells and extracellular vesicles across clinical trials and experimental assays.

Senior Data Integration Analyst, Children's Hospital of Philadelphia (CHOP) | 2010 – 2017

Developed tools for mitochondrial and exome variant analysis and ChIP-Seq / RNA-Seq reproducible reports. Studied miRNA differential expression and cleavage in retinoblastoma. Designed sequence manifest validation for the Pediatric Cardiac Genetic Consortium exome data hub. Developed a fork of SnpEff for HGVS nomenclature and myBiC, a portal for bioinformatics deliverables. Led the CHOP team building the GRIN epilepsy analysis portal and a Jupyter-based variant discovery platform.

Senior Research Associate, DuPont Crop Genetics | 2007 – 2010

Developed bioinformatics tools for transcriptome assembly analysis, miRNA target scanning, and gene annotation. Built LIMS systems for high-throughput mutagenesis screens.

Bioinformatics Programmer, Univ. of Pennsylvania, Bushman Lab | 2004 – 2007

Developed a bioinformatics pipeline for annotation and statistical analysis of HIV integration sites, microbial diversity, and viral resistance mutations.

SKILLS

Product & Leadership: Product Strategy · Roadmap Planning · Feature Prioritization · OKRs · PRDs & User Stories · GTM Strategy · Stakeholder Management · Customer Interviews · Market Research · Product Analytics · Technical Sales · Technical Leadership · Agile/Scrum (Linear, Jira, Shortcut)

Genomics & Bioinformatics: Population Genomics · GWAS · NGS · RNA-Seq · ChIP-Seq · ATAC-Seq · Variant Calling & Annotation · WES/WGS Analysis · Ontologies · Alignment & Annotation Pipelines · Data Pipelines · LIMS · Flow Cytometry · Reproducible Research · Drug Discovery · Sequence Analysis

Software & Cloud: Software Architecture · Scientific Computing · AI Agents · Cloud Infrastructure (AWS, GCP, Azure) · Kubernetes · Terraform · Docker/Singularity · Containerization · RESTful APIs · Django/Flask/Node.js · R Shiny · PostgreSQL/MongoDB/MariaDB/TileDB · Pipeline Frameworks (Nextflow, nf-core, Snakemake, WDL, CWL)

Languages: R, Python, Perl, Java, Groovy, Scala, JavaScript & jQuery, C++, C, PHP

EDUCATION

PhD, Information Science — Drexel University
MS, Computer Science — North Carolina State University
BS, Biology — Wake Forest University

SELECTED PUBLICATIONS (H-INDEX 30)

Leipzig, J., Nüst, D., Hoyt, C.T., Ram, K., Greenberg, J. The role of metadata in reproducible computational research. *Cell Patterns* 2021, 100322.

Leipzig, J., Bakis, Y., Wang, X., Elhamod, M., Diamond, K., Dahdul, W., Karpatne, A., Maga, M., Mabee, P., Bart, H.L., et al. Biodiversity Image Quality Metadata Augments Convolutional Neural Network Classification of Fish Species. *Metadata and Semantics Research* 2020, 3–12.

Leipzig, J. Computational Pipelines and Workflows in Bioinformatics. In *Reference Module in Life Sciences*; Elsevier, 2018.

Leipzig, J. A review of bioinformatic pipeline frameworks. *Briefings in Bioinformatics* 2017. †

Leipzig, J., Li, X.-Y. Data Mashups in R. *O'Reilly*, 2009 (reissued 2011).

Leipzig, J., Pevzner, P., Heber, S. The Alternative Splicing Gallery (ASG): bridging the gap between genome and transcriptome. *Nucl. Acids Res.* 2004, 32: 3977–3983.

† ISI Top 1% Highly Cited

PRESENTATIONS & POSTERS

"Variant Warehousing," ecSeq Berlin Summer School 2026 · AWS Genomics Community Day 2025 · BeginNGS Spring Workshop 2025 · Drexel-CODATA FAIR-RRDM Workshop 2019 · CYTO 2018 · MidAtlantic Bioinformatics Conference 2016 · Mitochondria, Aging & Disease Symposium 2013 · IEEE BIBM 2012 · ASHG 2012 · RECOMB 2004 · Rocky Mountain Bioinformatics Meeting 2003

SOFTWARE & TUTORIALS

Rhylthyme (rhylthyme.com) — JSON-based markup language and toolchain for real-time programs that coordinate parallel tracks of work, with a CLI runner, web visualizer, and MCP server for AI-assisted scheduling.

MITOMASTER — web application enabling clinicians to investigate mitochondrial mutations in sequenced or genotyped samples; used worldwide.

myBiC — Django app for authentication and delivery of bioinformatics deliverables for core facilities.

awesome-reproducible-research — curated, community-driven collection of reproducible research case studies, projects, and tutorials.

TEACHING

Introduction to Data Science (INFO103) — Adjunct Instructor, Drexel University.

Programming for Biology — Instructor & Teaching Assistant, Cold Spring Harbor Laboratory.

PROFESSIONAL ASSOCIATIONS & CERTIFICATIONS

PhillyR — Founder & Organizer of the Philadelphia R user community. · Sun Certified Java Programmer.